

Heads at a distance

The dyadic machine of Expo 1 prices space by grade and movement by the tree metric, but it treats all heads alike: any number of them, each moving at unit cost. Real machines do not extend that courtesy. A pointer that stays hot lives in registers and low cache levels; keep many of them hot and they compete for the same fast silicon. This exposition adds that competition to D as a **priced, structural** resource — no cache simulator, no eviction model: one inequality and one surcharge.

1 The model

Every head i is mounted at a distance $d_i \in \mathbb{N}$. Two rules (cache-v0, after convo [55]–[59]):

1. **Budget.** An assignment is admissible iff $\sum_i 2^{-d_i} \leq 1$ — the Kraft inequality. Only so much fits nearby: head i occupies the fraction 2^{-d_i} of fast state, and the shares are disjoint (no overlap between heads, by design: overlap is what makes sharing models treacherous).
2. **Surcharge.** Every operation of head i costs $1 + d_i$ instead of 1: the distance is the head’s address depth in an implicit fast-state tree. A hot head is near and therefore cheaper to touch.

The plain machine is the fiber $d \equiv 0$:

LEAN $\vdash$ Adic.Dyadic.distCost_zero

Distance is a refinement

At distance zero everywhere, cost is exactly the old cost — every Expo 1 theorem survives as the $d \equiv 0$ fiber, and cost remains a homomorphism on words.

```
theorem distCost_zero {k : Nat} (word : ActionWord k) :
  distCost (fun x => 0) word = actionCost word
axioms: propext      hash: 86a107f6f54c
```

The Kraft condition is not an analogy but the region algebra itself: admissible distances are exactly antichains of dyadic mount slots.

LEAN $\vdash$ Adic.Dyadic.kraft_iff_mounting

Kraft = an antichain of mount slots

A distance assignment satisfies Kraft **iff** the heads can be mounted injectively on pairwise prefix-incomparable paths with $|\text{path}_i| = d_i$ — prefix-free codes, dyadic disjointness, and fast-state budgeting are one condition. Both directions, with a constructive canonical mounting.

```
theorem kraft_iff_mounting {k : Nat} (dists : Dists k) :
  Kraft0k dists  $\leftrightarrow$ 
   $\exists$  paths,
    ( $\forall$  (head : Fin k), (paths head).length = dists head)  $\wedge$ 
    Function.Injective paths  $\wedge$ 
     $\forall$  {left right : Fin k},
      left  $\neq$  right  $\rightarrow$  PrefixIncomparable (paths left) (paths right)
axioms: Classical.choice, Quot.sound, propext      hash: 9f2039740bce
```

2 Streaming survives distance

LEAN $\vdash$ Adic.Dyadic.distZipCost_closed

Zip at a distance, exact

The zip schedule of Expo 1 under fixed distances has an exact closed-form cost, still linear in 2^n — the surcharge enters as the multiplicative $(1 + d)$ and nothing worse. The convo’s own worked example ([59]) — three heads at uniform distance 2, Kraft mass $\frac{3}{4}$ — is instantiated and checked:

```
theorem distZipCost_closed (dists : Dists 3) (n : Nat) :
  distZipCost dists n =
    distZipLeafCost dists * 2 ^ n + distZipNodeCost dists * (2 ^ n - 1)
axioms: Quot.sound, propext      hash: f053f2677702
```

LEAN $\vdash$ Adic.Dyadic.uniformThreeDists_kraft

...and [59]’s example is admissible

$3 \cdot 2^{-2} \leq 1$, in the all-natural formulation, no real numbers anywhere.

```
theorem uniformThreeDists_kraft : Kraft0k uniformThreeDists
axioms: Quot.sound, propext      hash: 8e9b9e91f167
```

3 The entropy bound

Which distances should a program choose? Count the touches: head i touched c_i times out of n . Choosing distances against inverse empirical frequency is admissible and essentially optimal — the fast-state allocation problem **is** source coding.

LEAN $\vdash$ Adic.Dyadic.empiricalDists_kraft

Inverse-frequency distances are admissible

The inverse-frequency choice $d_i = \lceil \log_2(n/c_i) \rceil$ satisfies Kraft — the classical prefix-code argument, in $\mathbb{N}$.

```
theorem empiricalDists_kraft {k : Nat} (touches : List (Fin k))
  (hnonempty : 0 < touches.length)
  (hpositive :  $\forall$  (head : Fin k), 0 < touchCount touches head) :
  Kraft0k (empiricalDists touches)
axioms: Quot.sound, propext      hash: 29c28abfbd7
```

LEAN $\vdash$ Adic.Dyadic.touchCost_empirical_entropy_bound

Achievability: cost $\leq$ entropy + slack

Under those distances the total touch cost is bounded by the empirical entropy of the touch sequence (count form $\sum_i c_i \log(n/c_i)$) plus explicit linear slack — the $H(p) + O(1)$ per touch of convo [57], with every ceiling accounted.

```
theorem touchCost_empirical_entropy_bound {k : Nat} (touches : List (Fin k))
  (hnonempty : 0 < touches.length)
  (hpositive :  $\forall$  (head : Fin k), 0 < touchCount touches head) :
  touchCost (empiricalDists touches) touches  $\leq$ 
    empiricalFloorLogTotal touches + 2 * touches.length
axioms: Quot.sound, propext      hash: e49667460df8
```

LEAN $\vdash$ Adic.Dyadic.touchCost_entropy_lower

Optimality: no distances beat entropy

Conversely, **every** Kraft-admissible assignment pays at least the empirical entropy (floor form, explicit slack): Shannon’s lower bound, carried by a discrete Gibbs/log-sum inequality proved by the counting route —

```
theorem touchCost_entropy_lower {k : Nat} (dists : Dists k)
  (touches : List (Fin k)) (hKraft : Kraft0k dists) :
  touches.length + empiricalFloorLogTotal touches  $\leq$ 
    touchCost dists touches + touches.length
axioms: Quot.sound, propext      hash: 3e2e47c38273
```

LEAN $\vdash$ Adic.Dyadic.discrete_gibbs_log_sum

The discrete log-sum inequality

$\sum_i c_i \lceil \log_2(n/c_i) \rceil \leq \sum_i c_i d_i + n$ for any Kraft-admissible d — all-natural, zero-count heads included.

```
theorem discrete_gibbs_log_sum {k : Nat} (total : Nat)
  (counts : Fin k  $\rightarrow$  Nat) (dists : Dists k) (htotal : finSum counts = total)
  (hKraft : Kraft0k dists) :
  floorLogTotal total counts  $\leq$  distCountTotal counts dists + total
axioms: Quot.sound, propext      hash: 21aec34cbd41
```

Together: fast state allocated like a prefix code, priced like a code length, optimal like a Shannon code. Biased search trees (Bent–Sleator–Tarjan) are the data-structure ancestor; adaptive distance assignment via dynamic Huffman/FGK reweighting is a known road, deferred.

4 What this model refuses

DESK-PROVED — not mechanized

Kraft pricing is not cliff pricing

k -way merge under Kraft distances costs $n \log n$ for **every** k — the multi-way tradeoff cancels exactly ([57]’s honest computation). The external-memory win ($\log$ base $\frac{M}{B}$) comes from the capacity **cliff**, which this model deliberately lacks: it never under-charges, and pessimism is the right failure mode for a verifier. Mechanizing this cancellation is phase 3; until then it is desk-level.

UNFORMALIZED

The write-head distance ([72])

Should zip’s write head, moving twice the data, get a larger share? Touch-frequency pricing says no (all three heads touch equally often); data-rate pricing says yes. The tension is exactly touch-frequency distances vs size-aware distances, and its resolution belongs to the size-aware distance refinement (v1), where a head’s state size enters the budget.